

MILCOM 2027 — Tutorial TUT13
AI-Enabled Antenna-to-System Design with LLMs, MCP and MATLAB

Overview

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Agenda

Why AI for antenna design, analysis, and optimization

Challenges:

- Electromagnetic analysis is very time consuming, especially when repeated across a design space
- Exploring large parametric design spaces and selecting topologies is slow and expensive
- Measuring full 3D radiation patterns is costly; often only sparse 2D cuts are available
- Optimizing antennas with many parameters is computationally intractable
- Authoring designs and connecting components to system-level simulation is fragmented and manual
- Antenna designers might not be familiar with AI

Agenda

Three-part tutorial and optional optimization bonus

Solutions and benefits:

Part 1 — Agentic AI workflows

- Create antenna geometries and run engineering tools using LLMs, MCP, and domain skills.

Part 2 — AI models and pattern reconstruction

- Explore designs with pretrained models and AutoML; reconstruct 3D patterns from sparse cuts.

Part 3 — Antenna-to-system integration

- Use antenna, RF PCB, and pattern artifacts in end-to-end RF system simulations.

Bonus — Antenna optimization

- Explore surrogate-assisted search and design refinement after the main tutorial.

Agenda

Part 1 — Agentic AI workflows with LLMs, MCP, and skills

Part 1: Agentic AI Workflows

45 minutes • Opening Demo + Lecture

- LLM client workflow for antenna and RF PCB design authoring
- Model Context Protocol connection between the LLM interface and engineering tools
- LLM skills for producing and modifying structured PCB geometries
- Skills for RF metrics, MATLAB code optimization, pattern analysis, and design checking
- Natural-language intent to parametric geometry, simulation, metric extraction, and iteration
- Governance: reproducibility, validation, model boundaries, and human review

Agenda

Part 2 — AI models and pattern reconstruction (60 minutes)

Part 2A: Pretrained Antenna Models and AutoML

25 minutes • Lecture + Demo + Online Exercise

- Pretrained surrogate model library for common antenna families
- Rapid topology selection and design-space exploration using learned priors
- AutoML workflow for training custom predictors for new geometries and requirements
- Dataset generation from parametric EM simulations
- Model selection, validation, error analysis, and model-usage boundaries

Part 2B: AI-Based Pattern Reconstruction

25 minutes • Lecture + Demo + Online Exercise

- Reconstructing full 3D patterns from sparse azimuth and elevation cuts
- U-Net-based generative model architecture and training approach
- Accuracy and speed comparison against traditional reconstruction methods
- Validation across diverse antenna types; current limitations and emerging challenges

Break — 10 minutes

Agenda

Part 3 — System integration; bonus — optimization

Part 3: Antenna-to-System Integration

35 minutes • Demo

- Inserting antenna, RF PCB, and pattern artifacts into system-level simulations
- Linking EM results to RF chains, waveforms, and link-budget models
- RF transmitter and receiver chain examples using antenna artifacts
- End-to-end performance evaluation for design decisions
- How LLM/MCP workflows shorten the path from component design to system simulation

Bonus: Antenna Optimization

Optional 35 minutes • Lecture + Demo + Exercise

- Optimization across different antenna families; evolved pixelated topologies
- Surrogate-assisted search for rapid design-space exploration
- Customized genetic algorithm refinement for miniaturization and performance constraints
- Trade-offs among speed, accuracy, manufacturability, and EM fidelity
- Python and MATLAB co-design connecting AI, optimization, and EM simulation

Sign-up Form

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